Pricing Index-Linked Hybrids under a Local Volatility Market Model

Working notes on Pucci (2012b)

Reference paper: M. Pucci, Pricing Index-Linked Hybrids under a Local Volatility Market Model*, Banca IMI, Milan, February 22, 2012 (revised May 11, 2012). SSRN working paper no. 2056277. Available at* https://ssrn.com/abstract=2056277.

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1 Trade and Motivation

The target product is a *cash-settled swaption with strike contingent on an equity index level* – a hybrid European-style payoff combining a swap-rate fixing with an equity-index fixing, both observed at the swaption expiry T. The payoff is

$$\text{Payoff}_T \;=\; \widehat{A}_T \left(\theta(S_T - U_T)\right)^+, \qquad \theta \in \{+1, -1\}, \tag{1}$$

where $\widehat{A}_T$ is the cash annuity (defined below), S_T is the swap rate fixing and U_T is the index fixing at T. The strike of the embedded cash swaption is therefore the index level itself.

Modelling challenge. Pricing such a hybrid traditionally involves combining a stochastic short-rate model (or LMM) with an equity model (Black–Scholes / Heston), introducing several sources of trouble:

- *Unobservable factors*: short rate, stochastic-vol drivers, auxiliary state variables.
- *Hybrid correlation structure*: many cross-correlations between unobservable factors, with no liquid market quotes.
- *Smile fitting*: classical setups either miss the rate smile (BS+SR with equity vol surface, e.g. Brigo–Mercurio [1]) or the equity smile (DD-LMM only). Heston-LMM combinations (Grzelak–Oosterlee [2]) capture both but at the price of heavy machinery and indirect correlation handling.

Modelling goal. Build a framework that prices the hybrid directly in terms of *market observables*, with:

- full equity skew/smile via Dupire local volatility on the index forward;
- full rate skew/smile via Dupire local volatility on the convexity-adjusted forward swap rate;
- a single *exogenous* correlation ρ, specified directly on the two market-observable martingales – no implicit correlation through unobservable state variables.

2 Definitions

Let $T_0 = T < T_1 < \cdots < T_n$ be a time structure with year fractions $y_i = \text{yf}(T_{i-1}, T_i)$, D_{tT} the risk-free zero-coupon bond.

2.1 Forward swap rate and annuity

The forward swap rate $S_t = S_{tT}(T_1, \ldots, T_n)$ is defined in the standard way against the (true) annuity

$$A_t = A_t(T_1, \ldots, T_n) \;=\; \sum_{i=1}^{n} y_i \, D_{tT_i}, \tag{2}$$

and is a martingale under the A_t-numeraire (swaption) measure $\mathbb{Q}^A$.

2.2 Cash annuity

Definition 1 (Cash annuity)**.** *The cash annuity, used in cash-settled swaption conventions, is the flat-curve proxy*

$$\widehat{A}_T \;=\; \sum_{i=1}^{n} \frac{y_i}{(1 + y_i \, S_T)^{\,\mathrm{yf}(T,T_i)}}. \tag{3}$$

Each market discount D_{TT_i} is replaced by a synthetic discount $(1 + y_i S_T)^{-\mathrm{yf}(T,T_i)}$ derived under a flat-curve assumption with yield S_T. Crucially, $\widehat{A}_T$ is a deterministic function of S_T alone – this is what makes the framework tractable.

2.3 Index forward

For the equity index U_t (spot), the T-forward price is

$$U_{tT} \;=\; \mathbb{E}_t^T[\,U_T\,] \;=\; U_t \, \frac{1 - \mathrm{div}_t(T)}{D_{tT}} \cdot e^{-r^{\mathrm{repo}}(T-t)}, \tag{4}$$

or, equivalently, U_{tT} is the unique martingale under the T-forward measure $\mathbb{Q}^T$ such that $U_{TT} = U_T$. (The exact formula above is illustrative; what matters is that U_{tT} is a $\mathbb{Q}^T$-martingale by construction.)

2.4 The hybrid payoff

The index-linked cash-settled swaption pays (1) at T. Note:

- It is a European payoff: only the joint distribution of (S_T, U_T) at T matters, not the path.
- The cash annuity $\widehat{A}_T$ is a function of S_T alone, hence of the pair only through the swap-rate component.
- For $\theta = +1$, the payer of the swap receives a positive cash flow when S_T exceeds the index strike U_T.

3 Pricing Framework

3.1 Generic numeraire

For any numeraire N_t with associated measure $\mathbb{Q}^N$, the present value at $t \le T$ is

$$v_t \;=\; N_t \, \mathbb{E}_t^N \left[\frac{\widehat{A}_T \, (\theta(S_T - U_T))^+}{N_T} \right]. \tag{5}$$

The natural choices are $N_t = A_t$ (swaption measure – makes S_t martingale, but $\widehat{A}_T/A_T$ enters the integrand) or $N_t = D_{tT}$ (T-forward measure – the cleaner option).

3.2 Switch to the T-forward measure

Definition 2 (Convexity-adjusted forward swap rate)**.**

$$F_{tT} \;\equiv\; \mathbb{E}_t^T[\,S_T\,]. \tag{6}$$

At $t = T$, $F_{TT} = S_T$ trivially. The point of this definition is that F_{tT} is a $\mathbb{Q}^T$-martingale by construction, with terminal value S_T.

Proposition 1 (T-forward pricing formula)**.** *The price of the index-linked cash-settled swaption is*

$$\boxed{v_t \;=\; D_{tT}\,\mathbb{E}_t^T\Big[\widehat{A}_T\,\big(\theta(F_{TT} - U_{TT})\big)^+\Big]\,.} \tag{7}$$

Both F_{tT} and U_{tT} are $\mathbb{Q}^T$-martingales.

Proof. Setting $N_t = D_{tT}$ in (5) gives the $\mathbb{Q}^T$-expectation; $D_{TT} = 1$ removes the denominator. $F_{TT} = S_T$ by (6), $U_{TT} = U_T$ by definition of forward, hence the integrand is $\widehat{A}_T(\theta(F_{TT} - U_{TT}))^+$. F_{tT} is a $\mathbb{Q}^T$-martingale by construction; U_{tT} is a $\mathbb{Q}^T$-martingale because the T-forward of any tradable asset is (spot/D_{tT} adjusted for carry, which itself is a tradable combination). □

Remark 1 (Why the T-forward measure is preferred)**.** *Working under $\mathbb{Q}^A$ would force us to model the joint distribution of (S_T, U_T, A_T), including the ratio $\widehat{A}_T/A_T$ – itself a non-trivial function of the term structure. Under $\mathbb{Q}^T$, only the* joint marginal *(F_{TT}, U_{TT}) at T matters, since $\widehat{A}_T$ is a function of $F_{TT} = S_T$ alone. We have moved from a 3-dimensional modelling problem (rate, index, annuity ratio) to a 2-dimensional one (rate, index). This is the central simplification.*

4 Local Volatility Market Model

4.1 Model assumption

Assumption 1 (Local-volatility market model)**.** *F_{tT} and U_{tT} are local-volatility martingales under $\mathbb{Q}^T$:*

$$\frac{dF_{tT}}{F_{tT}} = \sigma_F(t, F_{tT})\,d(W^F)_t^T, \tag{8}$$

$$\frac{dU_{tT}}{U_{tT}} = \sigma_U(t, U_{tT})\,d(W^U)_t^T, \tag{9}$$

with constant instantaneous correlation

$$d\big\langle (W^F)^T, (W^U)^T \big\rangle_t \;=\; \rho\,dt. \tag{10}$$

Three ingredients. The model has *exactly*:

1. a local-vol surface $\sigma_F(t, \cdot)$ for the rate component;
2. a local-vol surface $\sigma_U(t, \cdot)$ for the index component;
3. a single scalar correlation ρ.

No short-rate dynamics, no auxiliary stochastic-vol factor, no hidden state variable.

4.2 Why "market model"?

The two state variables are quantities that map directly to market quotes:

- **Convexity-adjusted CMS rate** F_{0T}: implied from CMS spread quotes, or built via static replication on top of a SABR-fitted swaption surface. Many interest-rate desks already maintain a CMS interpolator producing this number directly.
- **Index forward** U_{0T}: implied from put–call parity on liquid index options, or from spot/dividend/repo discount-curve construction.

This is in contrast to "factor models" where the underlying state variables (short rate, stochastic-vol process) cannot be quoted by market makers.

5 Calibration to Vanilla Markets

The strength of the framework is that all three ingredients calibrate *independently* against listed European instruments, all expressed as $\mathbb{Q}^T$-expectations.

5.1 Vanilla cash-settled swaption

Proposition 2 (Cash swaption price)**.** *The strike-K cash-settled swaption (vanilla) is priced as*

$$\text{CashSwaption}_t(K) \;=\; D_{tT}\,\mathbb{E}_t^T\Big[\widehat{A}_T\,\big(\theta(F_{TT}-K)\big)^+\Big]\,. \tag{11}$$

Proof. Set $U_T = K$ (deterministic) in (7). □

This is the natural calibration target for $\sigma_F(t,\cdot)$: the smile of F_{tT} across strikes K matches the broker-quoted cash-settled swaption smile (or, equivalently, the SABR-interpolated swaption surface).

5.2 CMS cap/floor

Proposition 3 (CMS cap/floor price)**.** *The strike-K CMS cap/floor (settling at T) is*

$$\text{CMSCapFloor}_t(K) \;=\; D_{tT}\,\mathbb{E}_t^T\Big[\big(\theta(F_{TT}-K)\big)^+\Big]\,. \tag{12}$$

Proof. Same as Proposition 2 but with $\widehat{A}_T$ replaced by 1, since CMS cap/floor pays the rate without the annuity weighting. □

This appears simpler – no $\widehat{A}_T$ factor – but CMS caps/floors are not generally broker-listed, so the cash-swaption route is preferred in practice.

5.3 Vanilla index option

Proposition 4 (Index option price)**.**

$$\text{IndexOption}_t(K) \;=\; D_{tT}\,\mathbb{E}_t^T\Big[\big(\theta(U_{TT}-K)\big)^+\Big]\,. \tag{13}$$

This is the standard Black-style formula on the index forward; the local-vol surface $\sigma_U(t,\cdot)$ is calibrated to the listed index option smile via the Dupire formula.

5.4 Calibration recipe

The model parameters are obtained in four sequential steps:

1. **Set F_{0T}.** Use the broker CMS quote at tenor T (typically derived through a SABR-based CMS interpolator). This fixes the level of the rate component.

2. **Calibrate $\sigma_F(t, \cdot)$.** Either:

   - *from cash swaptions:* apply Dupire on the F_{tT} martingale using the smile defined by (11); or
   - *from CMS caps/floors:* apply Dupire on F_{tT} using (12) (cleaner formula but instruments are usually not listed).

3. **Calibrate $\sigma_U(t, \cdot)$.** Apply Dupire on the U_{tT} martingale using the listed index option smile via (13). This is the standard Dupire calibration for an equity forward.

4. **Specify ρ.** Estimate exogenously, since no liquid instrument quotes a CMS–equity correlation directly. Two practical proxies:

   - Compute historical F from a curve-interpolated forward swap rate (or, more accurately, augment with a Black ATM convexity adjustment), take its log-return correlation with the index forward;
   - Use OTC structured-product back-out of ρ from past trades (if data is available).

   The exogenous nature of ρ is a feature, not a bug: it isolates the correlation parameter from the smile-fitting process.

5.5 Self-consistency checks

After calibration, three identities must hold to numerical precision under the calibrated model:

$$\mathbb{E}_0^T[F_{TT}] = F_{0T}, \tag{14}$$

$$\mathbb{E}_0^T[U_{TT}] = U_{0T}, \tag{15}$$

$$(11) \text{ at all } K = \text{market cash-swaption smile}, \tag{16}$$

plus the analogous identity for index options. These are the calibration's necessary smoke tests.

6 Pricing the Hybrid

6.1 Joint distribution under $\mathbb{Q}^T$

Under Assumption 1, the joint density of (F_{TT}, U_{TT}) under $\mathbb{Q}^T$ is determined by:

- the marginal of F_{TT} (set by σ_F);
- the marginal of U_{TT} (set by σ_U);
- the constant correlation ρ between the driving Brownians.

The marginals match the vanilla markets exactly by construction; the joint structure is determined by ρ.

6.2 Hybrid price

By Proposition 1,

$$v_0 \;=\; D_{0T} \iint_{\mathbb{R}_+^2} \widehat{A}_T(s)\,\big(\theta(s-u)\big)^+\, p_{F,U}(s,u)\,ds\,du, \tag{17}$$

where $p_{F,U}(s,u)$ is the joint density of (F_{TT}, U_{TT}) under $\mathbb{Q}^T$. In practice this is computed via:

- *2D Monte Carlo* of the SDE pair (8)–(9) with two correlated Brownians, discretised by Euler-on-log or Andersen QE;
- *2D quadrature* on the joint density when one prefers a deterministic approach;
- *Conditional 1D pricing* when one of the two marginals is analytically tractable (e.g. in the no-smile Black limit – see below).

7 Special Cases and Sanity Limits

7.1 No-smile (Black) limit

Take $\sigma_F(t,\cdot) = \sigma_F$ and $\sigma_U(t,\cdot) = \sigma_U$ both constant. Then F_{TT} and U_{TT} are jointly lognormal under $\mathbb{Q}^T$ with

$$\log F_{TT} \sim \mathcal{N}\big(\log F_{0T} - \tfrac{1}{2}\sigma_F^2 T,\ \sigma_F^2 T\big)\,, \tag{18}$$

$$\log U_{TT} \sim \mathcal{N}\big(\log U_{0T} - \tfrac{1}{2}\sigma_U^2 T,\ \sigma_U^2 T\big)\,, \tag{19}$$

$$\mathrm{Corr}(\log F_{TT}, \log U_{TT}) \;=\; \rho. \tag{20}$$

In this regime the hybrid price reduces to a 2D integral against a bivariate lognormal density.

7.2 $\rho = 0$ decomposition

Proposition 5 (Independence reduction)**.** *If $\rho = 0$, the hybrid price reduces to a 1-D integral:*

$$v_0\big|_{\rho=0} \;=\; D_{0T} \int_{\mathbb{R}_+} \mathrm{CashSwaption}_0(K=u)/D_{0T} \;\cdot\; p_U(u)\,du, \tag{21}$$

where p_U is the marginal density of U_{TT} under $\mathbb{Q}^T$.

Proof. Conditioning on $U_{TT} = u$ (which is independent of F_{TT} when $\rho = 0$):

$$\begin{aligned}\mathbb{E}_0^T\Big[\widehat{A}_T(\theta(F_{TT}-U_{TT}))^+\Big] &= \int \mathbb{E}_0^T\Big[\widehat{A}_T(\theta(F_{TT}-u))^+ \,\Big|\, U_{TT}=u\Big]\; p_U(u)\,du \\ &= \int \mathbb{E}_0^T\Big[\widehat{A}_T(\theta(F_{TT}-u))^+\Big]\; p_U(u)\,du,\end{aligned}$$

which is the 1-D integral of the vanilla cash-swaption price (at strike u) against the index density. □

This is an excellent regression test: with ρ set to zero, the hybrid pricer must equal a 1-D integral over the (already calibrated) vanilla cash-swaption surface.

7.3 $\rho = \pm 1$ degeneracy

When $\rho = +1$, the two driving Brownians coincide and U_{TT} becomes a deterministic monotone function of F_{TT} (specifically, the function $\Phi_U^{-1} \circ \Phi_F$ via percentile mapping). The joint becomes degenerate (supported on a 1-D curve), and the hybrid price reduces to a 1-D integral against the marginal of F_{TT}. For $\rho = -1$ the same holds with the inverse percentile mapping. These are not realistic operating regimes but are useful boundary tests.

8 Generalisation to Other Hybrid Payoffs

The framework extends, without modification of the calibration steps, to any *European multi-asset hybrid* where one leg is a cash-swaption-like annuity and the other is a martingale on the T-forward measure. Replacing U with another $\mathbb{Q}^T$-martingale gives:

- **FX-linked cash swaption:** $U \to$ FX forward; σ_U from FX option market.
- **Commodity-linked cash swaption:** $U \to$ commodity forward; σ_U from commodity option smile.
- **Basket-linked cash swaption:** $U \to$ basket forward (after suitable martingale adjustment for the basket weights).

In each case the same three ingredients $(\sigma_F, \sigma_U, \rho)$ suffice and are calibrated as in Section 5.

9 Validation Notes for the Pricing Library

9.1 Building blocks

1. **Cash annuity.** Implement (3) with exact day-count conventions. Smoke test: for an annual schedule with $y_i = 1$ and $\mathrm{yf}(T, T_i) = i$, $\widehat{A}_T = (1 - (1 + S_T)^{-n})/S_T$ – the classical closed form.
2. **Index forward.** Build U_{0T} from spot, dividend schedule, repo rate and discount curve via (4). Cross-check with put-call parity: at any strike K, $C(K) - P(K) = D_{0T}(U_{0T} - K)$.
3. **Convexity-adjusted CMS rate.** Implement either the SABR-based CMS interpolator route, or static replication on the swaption surface; both must agree to a few basis points at a sensible strike grid.

9.2 Calibration self-checks

4. **Martingale property of F.** After calibrating σ_F, Monte Carlo $\mathbb{E}_0^T[F_{TT}]$ should equal the input F_{0T} to within MC error – if not, the local-vol calibration is mis-specified.
5. **Martingale property of U.** Same test for the index forward: $\mathbb{E}_0^T[U_{TT}] = U_{0T}$.
6. **Smile reproduction (rate).** Price strikes across the cash swaption smile via (11) under the calibrated σ_F; compare with the input market smile – this is the calibration self-check.
7. **Smile reproduction (index).** Same for index options via (13).

9.3 Hybrid pricing

8. **No-smile / no-correlation limit.** With both σ_F and σ_U constant and $\rho = 0$, validate the 2-D MC implementation against a closed-form integral over the bivariate-lognormal density.
9. **$\rho = 0$ decomposition.** Use Proposition 5: the hybrid price at $\rho = 0$ must equal the 1-D integral of the vanilla cash-swaption price against the index marginal density. The right-hand side uses only the already-calibrated vanilla pricer.
10. **$\rho = \pm 1$ degeneracy.** Set $\rho = 1$, MC should collapse on the percentile mapping curve. Confirms numerical stability at boundary.
11. **Sign monotonicity.** The hybrid price must be monotone in ρ when $\theta = +1$ (more positive correlation $\Rightarrow$ more in-the-money payoff for S_T moving with U_T). Verify across the ρ grid.

9.4 Vanilla price decomposition (advanced)

12. **Static-replication consistency.** If your library prices CMS caps/floors via static replication, equation (12) should match – proof of consistency between the swaption-route and the CMS-route calibrations of σ_F.

13. **Annuity weighting effect.** The ratio (11)/(12) at the same K measures the Annuity weighting effect; it should be roughly A_0/D_{0T} in the no-smile regime and approach the cash-PV01 in skewed regimes.

10 Practical Implementation Notes

10.1 Local volatility surface construction

The Dupire formula

$$\sigma_F(t,K)^2 \;=\; \frac{\partial_t C(t,K) + (\dots)}{\frac{1}{2}K^2\,\partial_K^2 C(t,K)} \tag{22}$$

needs care at the wings – the implied volatility surface beyond the listed strikes is typically extrapolated via SABR or a smile-arbitrage-free parametric form (SVI). The same applies to σ_U.

10.2 Two-dimensional Monte Carlo

The 2D simulation under $\mathbb{Q}^T$ is straightforward:

1. Discretise $[0,T]$ into N steps with $\Delta t = T/N$.
2. At each step, draw (Z_1, Z_2) standard normal with Corr $= \rho$ via Cholesky $Z_2 = \rho Z_1 + \sqrt{1-\rho^2}\,Z_3$.
3. Update via Euler-on-log: $F_{t+\Delta t} = F_t \exp\big[-\frac{1}{2}\sigma_F(t,F_t)^2\Delta t + \sigma_F(t,F_t)\sqrt{\Delta t}Z_1\big]$, similarly for U.

For high accuracy at long expiries or steep smiles, replace Euler with Andersen QE or moment-matched schemes.

10.3 Performance: 2D quadrature alternative

For European-only payoffs, a 2D quadrature on the calibrated bivariate density (e.g. Gauss-Legendre on a truncated lognormal grid, with copula-glued marginals) can be markedly faster than MC, especially for risk-sensitivity computation.

11 Connection to Other Pucci Papers

The methodology shares structural DNA with the other three Pucci papers in this family:

- **CMASW (Pucci 2012a)** also uses the swaption measure and a two-underlying martingale framework, but with the asset swap spread R^{asw} playing the role of U – and a Hagan-style linear swap-rate model in lieu of full local volatility.
- **CMT (Pucci 2014)** extends the linear swap-rate model to risky underlyings via credit-risk-adjusted discount and annuity, in the spirit of the same observable-rates philosophy.
- **T-Lock (Pucci 2019)** specialises to a deterministic-yield framework where the proxy-forward booking is shown to be gamma-conservative.

Reference. Pucci, M. (2012b). *Pricing Index-Linked Hybrids under a Local Volatility Market Model.* Working paper, Banca IMI, Milan. SSRN no. 2056277.

Companion papers. M. Pucci (2012a), *Constant Maturity Asset Swap Convexity Correction*, SSRN 1961545; M. Pucci (2014), *Constant Maturity Treasury Convexity Correction*, IJTAF 17(8), SSRN 3387961; M. Pucci (2019), *Hedging the Treasury Lock*, SSRN 3386521.

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[2] L. Grzelak and K. Oosterlee. *An Equity-Interest Rate Hybrid Model With Stochastic Volatility and the Interest Rate Smile.* 2010.